

Nafisat Ozohu Abdulsalam

Mechatronics Engineering | Medical imaging | Robotics

nafisatabdulsalam05@gmail.com

EDUCATION

Federal University of Technology, Minna (FUTMinna)

B.Eng. in Mechatronics Engineering | May 2023 - Feb. 2028 (expected)

Current CGPA: 4.61/5.00 | First-class standing

Relevant courses: Control Engineering I & II, Introduction to Robotics, Mechatronics System Design, and Image Processing.

SELECTED PUBLICATIONS

Beyond Dice: Evaluating Reliability and Robustness in Breast Ultrasound Lesion Segmentation Under Progressive Speckle Corruption

Contribution: breast ultrasound data preparation and manuscript development.

Accepted | MIRASOL · MICCAI 2026

Building MRI Capacity in Nigeria: Preliminary Report of the First Onsite Assembly of a Reusable, Low-Field MRI Scanner and Robotic Field-Mapping System in Nigeria

Nafisat Abdulsalam, et al.

Accepted | ESMRMB 2026

PHYSICIAN: A Deterministic Middleware Kernel for Mitigating Kinematic Hallucinations in Mobile Locomotion and Manipulation Domains

A. H. A. Showunmi, N. O. Abdulsalam, A. O. Asimiyu

Accepted | IEEE IRAI 2026

RESEARCH EXPERIENCE

Student Intern

Medical Artificial Intelligence Lab (MAI Lab) · Lagos, Nigeria | May - Oct. 2026

- Deployed and customised OpenMRS, contributing to frontend development, system configuration, and deployment for clinical use.
- Working on medical image analysis through the AutoPET project.

IMAGINE Summer School Participant

CAMERA MRI | July - Aug. 2026

- Led the assembly and characterisation of a reusable low-field MRI scanner and robotic three-axis field-mapping system.
- Reduced peak-to-peak field inhomogeneity by approximately 50% through three mapping-optimisation cycles; led abstract writing for work accepted at ESMRMB 2026.

SPARK Academy Participant

MICCAI-Endorsed Academy for African Medical Imaging Translation | Feb. - June 2026

- Preprocessed breast ultrasound imaging data for segmentation training and robustness evaluation in "Beyond Dice".
- Contributed to manuscript development and scientific writing for the study accepted at MIRASOL.

WORK EXPERIENCE

Editorial Intern

Nigerian Journal of Technological Development · Ilorin, Nigeria | Oct. 2025 - Present

- Performed first-level editorial review of 500+ manuscripts for journal formatting and submission standards.
- Assigned reviewers and monitored the peer-review process through the manuscript management system.

Engineering Intern

Autokontrol Technology · Ilorin, Nigeria | Oct. - Nov. 2025

- Designed PCB schematics and layouts, selected components, prepared boards for fabrication, and performed soldering and hardware testing for a high-voltage detection system.
- Supported installation, wiring, and testing of solar photovoltaic systems, including inverters, batteries, and charge controllers.

SELECTED PROJECTS

MRI & Mapping Robot | Aug. 2026

Assembly and characterisation of a reusable low-field MRI scanner, supported by a robotic three-axis field-mapping system.

AutoPET

Medical image analysis research through the AutoPET project at MAI Lab.

Surgical Intelligence | Aug. - Oct. 2026

A project exploring surgical intelligence.

OpenMRS Customisation | May - Oct. 2026

Adaptation of an open-source electronic medical record system for clinical use.

LEADERSHIP & COMMUNITY SERVICE

Campus Lead · Cowrywise FUTMinna Ambassadors | 2024-2026

Led campus activities promoting financial literacy and savings.

Build Lab Lead · IMAGINE Summer School, Team Nigeria | 2026

Coordinated hands-on activities in low-field MRI assembly and robotic field mapping.

Treasurer & Assistant General Secretary · NAMETS FUTMinna | 2023-2025

Managed financial records and supported meeting documentation and administrative coordination.

IEEE FUTMinna | 2023-2025

Participated in engineering activities and professional development programmes.

MSSN FUTMinna | 2023-2025

Participated in student community activities and religious programmes.

AWARDS & SCHOLARSHIPS

Petroleum Trust Development Fund (PTDF) Scholar | 2023-2028

VC's List for Academic Excellence | 2023-2028

1st Prize · Hult Prize FUTMinna | 2026

2nd Runner-up · The Republic Student Writing Contest | 2025

Solomon and Alice Foundation Scholar · ₦100,000 bursary | 2025

Halymah Ibrahim Babangida Foundation Scholar · ₦100,000 bursary | 2025-2026

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

AI in Medicine · DeepLearning.AI, Coursera | 2026

AI methods for medical diagnosis, prognosis, and treatment.

Data Engineering · IBM, Coursera | 2024-2025

Data pipelines, databases, and data processing with Python and SQL.

MATLAB and Simulink · MathWorks | 2025

Numerical computing, data visualisation, and system modelling and simulation.

Google Project Management Certification | 2026

Project planning, Agile methodologies, risk management, and stakeholder communication.

PROFESSIONAL MEMBERSHIPS & AFFILIATIONS

Institute of Electrical and Electronics Engineers (IEEE) | 2024-2026

National Association of Mechatronics Engineering Students | 2023-2028

Women in Machine Learning (WiML) | 2026